

B2 Symmetry and Relativity, 2021 — Model Solutions

Question 1 — Charges in a uniform electric field

Problem. Electron at $x = 0$, positron at $x = +d$, both at rest in S . At $t = 0$ a constant field $\mathbf{E}$ is switched on. (a) world lines for $\mathbf{E} = E\hat{\mathbf{x}}$; (b) motion in S' moving at $v\hat{\mathbf{x}}$ and momentum status; (c) for $\mathbf{E} = -E\hat{\mathbf{x}}$ with $d = 20$ cm, $E = 50$ kV/cm, find KE at collision, $N_{\min}$, photon λ ; (d) minimum d to produce W^+W^- , and linear vs. circular colliders.

(a)

Forces: electron feels $-eE\hat{\mathbf{x}}$, positron $+eE\hat{\mathbf{x}}$. They accelerate apart.

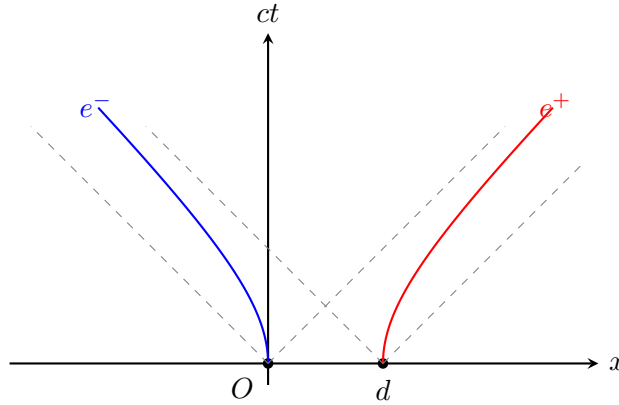
Equation of motion: $dp/dt = qE$ with $p = \gamma mv$:

$$p(t) = qEt.$$

Use $E_{\text{tot}}^2 = p^2c^2 + m^2c^4$ and integrate $v = pc^2/E_{\text{tot}}$:

$$x(t) = x_0 + \frac{c^2}{a_0} \left[\sqrt{1 + (a_0 t/c)^2} - 1 \right], \quad a_0 = eE/m_e. \quad (1)$$

Each world line is a hyperbola: vertical tangent at $t = 0$, asymptote at slope $\pm 1/c$. Mirror-symmetric about $x = d/2$.



Mirror hyperbolae about $x = d/2$, $a_0 = eE/m_e$.

(b)

In S' the surfaces $t' = \text{const}$ tilt: events “field on at electron” (origin) and “field on at positron” $(d, 0)$ are not simultaneous in S' . Time gap:

$$\Delta t' = -\gamma vd/c^2.$$

For $v > 0$ the positron's switch-on lies at *earlier* t' . So in S' the positron accelerates first while the electron remains at rest.

Trajectories remain hyperbolic (boost along $\hat{\mathbf{x}}$ maps hyperbolic motion to hyperbolic motion) but no longer mirror-symmetric.

Total 3-momentum in S : forces equal and opposite:

$$\dot{\mathbf{P}}_{\text{tot}} = -eE\hat{\mathbf{x}} + eE\hat{\mathbf{x}} = 0.$$

Hence $\mathbf{P}_{\text{tot}}$ conserved in S .

In S' : $E'_x = E_x$, but during the simultaneity-mismatch interval only one particle accelerates:

$$\frac{d\mathbf{P}'_{\text{tot}}}{dt'} = +eE\hat{\mathbf{x}}' \quad (\text{transient, positron only}).$$

Hence $\mathbf{P}'_{\text{tot}}$ *not* conserved during the transient: external (field) source is non-simultaneous in S' , so the system is open.

4-vector status. Each $\mathbf{P}_i$ is a 4-vector. The sum $\sum_i \mathbf{P}_i(t)$ depends on the spacelike slice ($t = \text{const}$ in S vs. $t' = \text{const}$ in S'). For an isolated system the slice-dependence drops out and the total is a 4-vector; for our open system (field on, particles still in field), the slice matters and $\mathbf{P}_{\text{tot}}$ is not a 4-vector.

$\mathbf{P}_{\text{tot}}$ conserved in S ; not in S' during transient; $\mathbf{P}_{\text{tot}}$ a 4-vector only for closed system.

(c)

Reverse field: electron pushed $+\hat{\mathbf{x}}$, positron $-\hat{\mathbf{x}}$. They meet at $x = d/2$ by symmetry. Each crosses distance $d/2$.

Work-energy theorem:

$$T = eE \frac{d}{2}.$$

Substitute $E = 5 \times 10^6 \text{ V/m}$, $d/2 = 0.10 \text{ m}$:

$$T = e \times 5 \times 10^6 \text{ V/m} \times 0.10 \text{ m} = 5 \times 10^5 \text{ eV}.$$

Hence $T = 0.500 \text{ MeV}$ per particle.

Total energy in lab (= ZMF by symmetry):

$$E_{\text{tot}} = 2(m_e c^2 + T) = 2(0.511 \text{ MeV} + 0.500 \text{ MeV}) = 2.022 \text{ MeV}.$$

Photon number. ZMF total $\mathbf{p} = 0$. Single photon has $|\mathbf{p}_\gamma| = E_\gamma/c \neq 0$: forbidden. Two photons back-to-back conserve both energy and momentum: $N_{\text{min}} = 2$.

Wavelength. Each photon carries $E_\gamma = E_{\text{tot}}/2$:

$$E_\gamma = 1.011 \text{ MeV}.$$

Apply $\lambda = hc/E_\gamma$ with $hc = 1240 \text{ eV nm}$:

$$\lambda = \frac{1240 \text{ eV nm}}{1.011 \times 10^6 \text{ eV}}.$$

Evaluate: $\lambda \approx 1.23 \text{ pm}$.

(d)

Threshold for $e^+e^- \rightarrow W^+W^-$: $E_{\text{tot}} \geq 2m_W c^2$:

$$2(m_e c^2 + T_{\text{min}}) = 2m_W c^2.$$

Neglect $m_e c^2 \ll m_W c^2$:

$$T_{\text{min}} = m_W c^2 = 80.379 \text{ GeV}.$$

Use $T_{\text{min}} = eE d_{\text{min}}/2$:

$$d_{\text{min}} = \frac{2m_W c^2}{eE}.$$

Substitute $eE = 5 \times 10^6 \text{ eV/m}$:

$$d_{\text{min}} = \frac{2 \times 80.379 \times 10^9 \text{ eV}}{5 \times 10^6 \text{ eV/m}}.$$

Evaluate: $d_{\text{min}} \approx 32 \text{ km}$.

Linear vs. circular. Synchrotron loss per turn $\propto \gamma^4/r$; for electrons at TeV scale $\gamma \sim 10^6$, prohibitive. Linear: no centripetal a , no synchrotron loss; ideal for $e^\pm$ at high energy. But single-pass: low luminosity. Circular: ideal for protons ($\gamma_p/\gamma_e = m_e/m_p$), continuous bunch reuse, high luminosity (LHC).

Question 2 — Liénard–Wiechert and synchrotron radiation

Problem. (a) Derive A^μ for an arbitrarily moving charge; explain radiative vs. non-radiative. (b) Synchrotron $r = 500 \text{ m}$, $B = 0.02 \text{ T}$: find γ , E , energy loss per turn (Larmor). (c) Detector at $+\hat{\mathbf{x}}$: show pulses, estimate duration.

(a)

Source 4-velocity $U^\mu = dX^\mu/d\tau$, with $U \cdot U = -c^2$. Retarded null separation $R^\mu = X_{\text{obs}}^\mu - X^\mu(\tau_{\text{ret}})$ with

$$R \cdot R = 0, \quad R^0 > 0.$$

Ansatz, manifestly covariant:

$$A^\mu = \frac{q}{4\pi\epsilon_0} \frac{U^\mu/c}{-R_\nu U^\nu}. \quad (2)$$

Verify in instantaneous rest frame at retarded event: $U^\mu = (c, \mathbf{0})$, $R^\mu = (r, \mathbf{r})$ with $r = |\mathbf{r}|$:

$$-R_\nu U^\nu = -(-rc) = rc.$$

Substitute:

$$A_{\text{rest}}^\mu = \frac{q}{4\pi\epsilon_0} \frac{(1, \mathbf{0})}{rc} \cdot c = \left(\frac{1}{c} \frac{q}{4\pi\epsilon_0 r}, \mathbf{0} \right).$$

Recognise static Coulomb. Both sides are 4-vectors; agreement in one frame $\Rightarrow$ all frames.

In a general frame, $U^\mu = \gamma(c, \mathbf{v})$, $-R_\nu U^\nu = \gamma cr(1 - \hat{\mathbf{r}} \cdot \mathbf{v}/c) \equiv \gamma cr\kappa$:

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r\kappa} \Big|_{\text{ret}}, \quad \mathbf{A} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v}}{r\kappa} \Big|_{\text{ret}}.$$

Field structure. $\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}$; chain rule through t_{ret} yields

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \left[\underbrace{\frac{(\hat{\mathbf{r}} - \mathbf{v}/c)(1 - v^2/c^2)}{\kappa^3 r^2}}_{\text{velocity, } \propto 1/r^2} + \underbrace{\frac{\hat{\mathbf{r}} \times [(\hat{\mathbf{r}} - \mathbf{v}/c) \times \dot{\mathbf{v}}/c]}{\kappa^3 r}}_{\text{acceleration, } \propto 1/r} \right]_{\text{ret}}.$$

Velocity term: $|\mathbf{S}| \propto 1/r^4$; flux through sphere $\rightarrow 0$ at infinity — non-radiative. Acceleration term: $|\mathbf{S}| \propto 1/r^2$; finite power at infinity — radiative. Only accelerating charges radiate.

(b)

Magnetic rigidity $p = eBr$:

$$pc = ceBr.$$

Use $ceB = 0.3 \text{ GeV}/(\text{c} \cdot \text{T} \cdot \text{m}) \cdot c$ practical constant: $pc[\text{GeV}] = 0.3 B[\text{T}] r[\text{m}]$:

$$pc = 0.3 \times 0.02 \times 500 \text{ GeV} = 3 \text{ GeV}.$$

Since $m_e c^2 = 0.511 \text{ MeV} \ll pc$:

$$E \approx pc = 3 \text{ GeV}.$$

$$\boxed{E \approx 3 \text{ GeV.}}$$

Lorentz factor:

$$\gamma = \frac{E}{m_e c^2} = \frac{3 \times 10^9 \text{ eV}}{0.511 \times 10^6 \text{ eV}}.$$

Evaluate: $\boxed{\gamma \approx 5.87 \times 10^3.}$

Larmor radiation. Proper acceleration for circular motion ($\mathbf{a} \perp \mathbf{v}$): $a_0 = \gamma^2 a$ with $a \approx c^2/r$:

$$a_0^2 = \gamma^4 c^4 / r^2.$$

Substitute into Larmor:

$$P_L = \frac{2}{3} \frac{e^2}{4\pi\epsilon_0} \frac{\gamma^4 c}{r^2}.$$

Numerics with $e^2/(4\pi\epsilon_0) = 2.31 \times 10^{-28} \text{ J m}$:

$$P_L = \frac{2}{3} \times 2.31 \times 10^{-28} \text{ J m} \times \frac{(5.87 \times 10^3)^4 \times 3 \times 10^8 \text{ m/s}}{(500 \text{ m})^2}.$$

Evaluate:

$$P_L \approx 2.2 \times 10^{-7} \text{ W}.$$

Period $T = 2\pi r/c$:

$$\Delta E = P_L T = \frac{e^2 \gamma^4}{3\epsilon_0 r}.$$

Numerics:

$$\Delta E = \frac{(1.602 \times 10^{-19} \text{ C})^2 (5.87 \times 10^3)^4}{3 \times 8.854 \times 10^{-12} \text{ F/m} \times 500 \text{ m}}.$$

Evaluate: $\boxed{\Delta E \approx 2.3 \times 10^{-15} \text{ J} \approx 14 \text{ keV per turn.}}$

(c)

Ultra-relativistic beaming: acceleration field concentrated in cone of half-angle

$$\Delta\theta \sim 1/\gamma.$$

Detector at $+\hat{\mathbf{x}}$ illuminated only while $\mathbf{v}$ within $\sim 1/\gamma$ of $\hat{\mathbf{x}}$. One pulse per revolution, period $T_{\text{rev}} = 2\pi r/c$.

Pulse duration. Emission arc subtends $\Delta\theta \sim 2/\gamma$ at the centre:

$$\Delta t_{\text{emit}} = \frac{r \cdot 2/\gamma}{v} \approx \frac{2r}{\gamma c}.$$

Retardation compression (source approaching detector at βc):

$$\Delta t_{\text{obs}} = (1 - \beta)\Delta t_{\text{emit}}.$$

Use $1 - \beta \approx 1/(2\gamma^2)$:

$$\Delta t_{\text{obs}} \approx \frac{2r}{\gamma c} \cdot \frac{1}{2\gamma^2} = \frac{r}{\gamma^3 c}.$$

Numerics:

$$\Delta t_{\text{obs}} = \frac{500 \text{ m}}{(5.87 \times 10^3)^3 \times 3 \times 10^8 \text{ m/s}}.$$

Evaluate: $\Delta t_{\text{obs}} \sim r/(\gamma^3 c) \approx 8 \text{ as.}$

Implies critical frequency $\omega_c \sim \gamma^3 c/r$, $\hbar\omega_c \sim 80 \text{ eV}$ (EUV/soft X-ray).

Question 3 — Crossed fields and the EM stress–energy tensor

Problem. (a) Electron from rest in $\mathbf{E} = (E, 0, 0)$, $\mathbf{B} = (0, B, 0)$: best frame and trajectory for $|E/B| > c$ and $|E/B| < c$. (b) Top row $T^{0\nu}$ from $T^{\mu\nu} = \epsilon_0 c^2 (-F^{\mu\lambda} F^\nu{}_\lambda - \frac{1}{4} g^{\mu\nu} F_{\kappa\lambda} F^{\kappa\lambda})$. (c) Capacitor moving along $\hat{\mathbf{x}}$: plates $\parallel xy$ then $\parallel yz$; $T^{\mu\nu}$ in rest and lab frames; energy flow.

(a)

Invariants:

$$\mathbf{E} \cdot \mathbf{B} = 0, \quad I_2 = E^2 - c^2 B^2.$$

$\mathbf{E} \perp \mathbf{B}$, so a frame exists in which one of $\mathbf{E}, \mathbf{B}$ is zero.

Case (i) $|E/B| > c$ ($I_2 > 0$): $\mathbf{E}$ wins. Boost along $\hat{\mathbf{z}}$ at speed

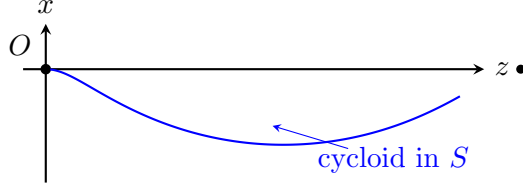
$$u = c^2 B/E < c$$

gives $\mathbf{B}' = 0$, pure $\mathbf{E}' = \sqrt{I_2} \hat{\mathbf{x}}$. In S' : hyperbolic motion along $\hat{\mathbf{x}}'$ ((1)). In S : same motion plus uniform drift $-u\hat{\mathbf{z}}$.

Case (ii) $|E/B| < c$ ($I_2 < 0$): $\mathbf{B}$ wins. $\mathbf{E} \times \mathbf{B}$ drift frame:

$$\mathbf{v}_{\text{drift}} = \frac{\mathbf{E} \times \mathbf{B}}{B^2} = (E/B)\hat{\mathbf{z}}, \quad |\mathbf{v}_{\text{drift}}| < c.$$

In S'' : $\mathbf{E}'' = 0$, pure $\mathbf{B}''$ along $\hat{\mathbf{y}}$. Electron at rest in S has velocity $-\mathbf{v}_{\text{drift}}$ in S'' , executes uniform circular motion in xz'' -plane. Boost back: cycloid in xz -plane, starting at a cusp (rest in S).



$$\boxed{|E/B| > c : \text{accelerated drift}; \quad |E/B| < c : \text{cycloid } (\mathbf{E} \times \mathbf{B} \text{ drift}).}$$

(b)

Use metric $\eta = \text{diag}(-1, +1, +1, +1)$. Faraday tensor:

$$F^{0i} = E^i/c, \quad F^{ij} = -\epsilon^{ijk} B_k.$$

Invariant scalar:

$$F_{\kappa\lambda} F^{\kappa\lambda} = 2(\mathbf{B}^2 - \mathbf{E}^2/c^2).$$

T^{00} . Compute $F^{0\lambda} F^0{}_{\lambda}$. Lower index: $F^0{}_i = -F^{0i} = -E^i/c$. So

$$F^{0\lambda} F^0{}_{\lambda} = (E^i/c)(-E^i/c) = -\mathbf{E}^2/c^2.$$

Substitute, using $g^{00} = -1$:

$$T^{00} = \epsilon_0 c^2 \left[\mathbf{E}^2/c^2 + \frac{1}{4}(-)(2)(\mathbf{B}^2 - \mathbf{E}^2/c^2) \right].$$

Wait — careful with signs: $-\frac{1}{4}g^{00}F_{\kappa\lambda}F^{\kappa\lambda} = -\frac{1}{4}(-1)(2)(\mathbf{B}^2 - \mathbf{E}^2/c^2) = \frac{1}{2}(\mathbf{B}^2 - \mathbf{E}^2/c^2)$:

$$T^{00} = \epsilon_0 c^2 \left[\mathbf{E}^2/c^2 + \frac{1}{2}(\mathbf{B}^2 - \mathbf{E}^2/c^2) \right].$$

Simplify:

$$T^{00} = \frac{1}{2}\epsilon_0 \mathbf{E}^2 + \frac{1}{2\mu_0} \mathbf{B}^2 \equiv u_{\text{EM}}.$$

T^{0i} . $g^{0i} = 0$, so

$$T^{0i} = -\epsilon_0 c^2 F^{0\lambda} F^i{}_{\lambda}.$$

Use $F^{0\lambda} = (0, \mathbf{E}/c)$ and lower index of $F^{i\mu}$. Take $i = x$:

$$F^{x\mu} = (E^x/c, 0, -B_z, B_y), \quad F^x{}_{\mu} = (-E^x/c, 0, -B_z, B_y).$$

Contract:

$$F^{0\lambda} F^x{}_{\lambda} = (E^x/c)(-E^x/c) \cdot [\text{but } F^{00} = 0] \dots$$

More directly: $F^{0\lambda} F^x{}_{\lambda} = (E^y/c)(-B_z) + (E^z/c)(B_y) = -(\mathbf{E} \times \mathbf{B})_x/c$.

Hence

$$T^{0i} = -\epsilon_0 c^2 \cdot (-(\mathbf{E} \times \mathbf{B})_i/c) = \epsilon_0 c (\mathbf{E} \times \mathbf{B})_i.$$

Recognise $\mathbf{S} = (\mathbf{E} \times \mathbf{B})/\mu_0$:

$$T^{0i} = \frac{(\mathbf{E} \times \mathbf{B})_i}{\mu_0 c} = \frac{S_i}{c}.$$

Combine:

$$\boxed{T^{0\nu} = (u_{\text{EM}}, \mathbf{S}/c) = \left(\frac{1}{2}\epsilon_0 \mathbf{E}^2 + \frac{1}{2\mu_0} \mathbf{B}^2, \frac{\mathbf{E} \times \mathbf{B}}{\mu_0 c} \right).} \quad (3)$$

T^{00} = EM energy density; T^{0i} = $c \times$ (momentum density) = energy flux/ c .

(c)

Plates $\parallel xy$, **motion along** $\hat{\mathbf{x}}$. Rest frame S^* : $\mathbf{E}^* = E\hat{\mathbf{z}}$, $\mathbf{B}^* = 0$. Faraday tensor non-zero entries: $F^{*03} = E/c$, $F^{*30} = -E/c$.

Compute $T^{*\mu\nu}$ from (3) extended; with $\mathbf{B}^* = 0$, energy density $u^* = \frac{1}{2}\epsilon_0 E^2$, $\mathbf{S}^* = 0$. Maxwell stress: tension along $\mathbf{E}$ (i.e. $\hat{\mathbf{z}}$), pressure transverse ($\hat{\mathbf{x}}, \hat{\mathbf{y}}$):

$$T_{(xy)}^{*\mu\nu} = u^* \text{diag}(+1, +1, +1, -1).$$

Boost to lab along $-\hat{\mathbf{x}}$ (i.e. lab moves at $-v\hat{\mathbf{x}}$ in S^* , equivalently S^* moves at $+v\hat{\mathbf{x}}$ relative to lab); $\Lambda^0_0 = \gamma$, $\Lambda^0_1 = \Lambda^1_0 = \beta\gamma$, $\Lambda^1_1 = \gamma$. Apply to the $(0, 1)$ block, which is $\text{diag}(u^*, u^*)$:

$$T'^{00} = \gamma^2(T^{*00} + \beta^2 T^{*11}) = \gamma^2 u^* (1 + \beta^2).$$

Off-diagonal:

$$T'^{01} = \gamma^2 \beta (T^{*00} + T^{*11}) = 2\gamma^2 \beta u^*.$$

And:

$$T'^{11} = \gamma^2 (\beta^2 T^{*00} + T^{*11}) = \gamma^2 u^* (1 + \beta^2).$$

Other components unchanged. Hence

$$T_{(xy), \text{lab}}^{\mu\nu} = u^* \begin{pmatrix} \gamma^2(1 + \beta^2) & 2\gamma^2\beta & 0 & 0 \\ 2\gamma^2\beta & \gamma^2(1 + \beta^2) & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad u^* = \frac{1}{2}\epsilon_0 E^2. \quad (4)$$

Plates $\parallel yz$, **motion along** $\hat{\mathbf{x}}$. Rest frame: $\mathbf{E}^* = E\hat{\mathbf{x}}$, $\mathbf{B}^* = 0$. Tension along $\hat{\mathbf{x}}$:

$$T_{(yz)}^{*\mu\nu} = u^* \text{diag}(+1, -1, +1, +1).$$

Boost along $\hat{\mathbf{x}}$: field $\parallel$ boost, so $\mathbf{E}, \mathbf{B}$ both unchanged in lab. Stress-energy tensor invariant under this boost:

$$T_{(yz), \text{lab}}^{\mu\nu} = u^* \text{diag}(+1, -1, +1, +1). \quad (5)$$

Energy flow. T^{0i} = energy flux (Poynting / c).

- (xy) : $T^{01} = 2\gamma^2 \beta u^* \neq 0$. Lab sees $\mathbf{B} = -\mathbf{v} \times \mathbf{E}/c^2$ (transverse to motion), so $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$ along $+\hat{\mathbf{x}}$. Electrostatic energy is convected with the moving capacitor.
- (yz) : $T^{0i} = 0$. Field $\parallel$ boost generates no $\mathbf{B}$, no $\mathbf{S}$. Energy density unchanged in lab; *no* EM energy flux despite the capacitor moving.

Convection only when boost has component $\perp \mathbf{E}$.

Question 4 — Symmetry, Lorentz group, rotation generators

Problem. (a) Define symmetry; identify SR's; SR-consistency criterion. (b) Show 2D rotation orthogonal, 1D boost not. (c) J_3 generates $R_z(\theta)$; meaning of $[J_i, J_j]$; rotations form a group. (d) Why boosts alone don't form a group; smallest group containing all 3D Lorentz transformations and connection to the rest.

(a)

A *symmetry* is a transformation of the dynamical variables that leaves the equations of motion form-invariant. By Noether: continuous symmetries $\leftrightarrow$ conservation laws.

SR symmetry. *Lorentz invariance:* laws form-invariant under $\Lambda^\mu{}_\nu$ (boosts + rotations); with translations forms the Poincaré group. Equivalently, $\eta_{\mu\nu}$ and c are invariant.

Criterion. An equation is SR-consistent iff both sides are Lorentz tensors of identical rank/index structure. Example: $\mathbf{F} = m\mathbf{A}$ is consistent (4-vectors); $\mathbf{F} = m\mathbf{a}$ alone is not (3-vector $\mathbf{F}$ does not transform as part of any single 4-vector without γ).

(b)

Clockwise 2D rotation:

$$R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Transpose:

$$R^T(\theta) = R(-\theta).$$

Multiply:

$$R^T R = R(-\theta)R(\theta) = R(0) = I.$$

Hence orthogonal.

1D Lorentz boost (rapidity η):

$$\Lambda(\eta) = \begin{pmatrix} \cosh \eta & -\sinh \eta \\ -\sinh \eta & \cosh \eta \end{pmatrix}.$$

Symmetric, so $\Lambda^T = \Lambda$:

$$\Lambda^T \Lambda = \Lambda^2 = \begin{pmatrix} \cosh 2\eta & -\sinh 2\eta \\ -\sinh 2\eta & \cosh 2\eta \end{pmatrix} \neq I.$$

Boost preserves Minkowski form, not Euclidean: $\Lambda^T \eta \Lambda = \eta$ (pseudo-orthogonal).

$R^T R = I; \Lambda^T \Lambda \neq I, \text{ but } \Lambda^T \eta \Lambda = \eta.$

(c)

J_3 **exponentiates.** Compute J_3^2 :

$$J_3^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \equiv P.$$

Note $J_3 P = J_3$, so $J_3^{2k} = P$, $J_3^{2k+1} = J_3$ ($k \geq 1$).

Expand exponential:

$$e^{-i\theta J_3} = (I - P) + P \sum_{k \geq 0} \frac{(-i\theta)^{2k}}{(2k)!} - iJ_3 \sum_{k \geq 0} \frac{(-i\theta)^{2k+1}}{(2k+1)!} \cdot (-i) \dots$$

Cleaner: separate even/odd:

$$e^{-i\theta J_3} = (I - P) + P \cos \theta - iJ_3 \sin \theta.$$

Use $-iJ_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$:

$$e^{-i\theta J_3} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = R_z(\theta).$$

Hence J_3 generates rotation about $\hat{\mathbf{z}}$.

Lie algebra.

$$[J_i, J_j] = i\epsilon_{ijk}J_k.$$

Meaning: structure constants of the rotation group; quantifies non-commutativity of infinitesimal rotations. Two infinitesimal rotations about $\hat{\mathbf{x}}$ then $\hat{\mathbf{y}}$ differ from the reverse order by a rotation about $\hat{\mathbf{z}}$.

Group axioms. Closure: BCH series

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]+\dots}$$

involves only nested commutators. By $[J_i, J_j] = i\epsilon_{ijk}J_k$, every commutator stays in $\text{span}\{J_k\}$; so $e^A e^B = e^{-i\theta'\hat{\mathbf{n}}'^k J_k}$, again a rotation. Identity: $R(0) = I$. Inverse: $R(-\theta, \hat{\mathbf{n}})$. Associativity: matrix multiplication. All four axioms hold.

Closure of $[J_i, J_j] \Leftrightarrow$ rotations form a group $\text{SO}(3)$.

(d)

Boost commutator:

$$[K_i, K_j] = -i\epsilon_{ijk}J_k.$$

Two boosts in different directions yield a *rotation* (Wigner / Thomas), not another boost. Hence $\text{span}\{K_i\}$ alone does not close under commutators: not a sub-algebra. By BCH, the product of two non-collinear boosts is a boost composed with a rotation; closure fails.

(Boosts in a fixed direction $\hat{\mathbf{n}}$, generated by a single K_i , do form a 1-parameter sub-group: rapidities add.)

Smallest group. Add rotations: full algebra $\{J_i, K_i\}$ obeys

$$[J_i, J_j] = i\epsilon_{ijk}J_k,$$

$$[J_i, K_j] = i\epsilon_{ijk}K_k,$$

$$[K_i, K_j] = -i\epsilon_{ijk}J_k.$$

This closes; exponentiation gives the proper orthochronous Lorentz group: $\boxed{\text{SO}^+(1, 3)}$.

Connection to the wider Lorentz group. $O(1, 3)$ has four disconnected components, classified by $\det \Lambda = \pm 1$ and $\Lambda^0_0 \gtrless 0$:

component	det	Λ^0_0	connector to SO^+
$\text{SO}^+(1, 3)$	+1	≥ 1	identity
$P \cdot \text{SO}^+$	-1	≥ 1	parity $P : (t, \mathbf{x}) \rightarrow (t, -\mathbf{x})$
$T \cdot \text{SO}^+$	-1	≤ -1	time reversal $T : (t, \mathbf{x}) \rightarrow (-t, \mathbf{x})$
$PT \cdot \text{SO}^+$	+1	≤ -1	PT

P, T are discrete, not reachable by exponentiating $\{J_i, K_i\}$; appended “by hand” to recover full $O(1, 3)$.